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Author(s): Jon Faust, John H. Rogers and Jonathan H. Wright

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JON FAUST
JOHN H. ROGERS
JONATHAN H. WRIGHT

News and Noise in G-7 GDP Announcements

Revisions to GDP announcements are known to be quite large in all G-7 countries; quarterly growth rate revisions are regularly more than a full percentage point at an annualized rate. We examine the predictability of these revisions using standard statistical tests of whether the preliminary announcement is a rational forecast of the subsequently revised data. Previous work suggests that U.S. GDP revisions are largely unpredictable, as would be the case if the revisions reflect news not available at the time that the preliminary number is produced. We find that the degree of predictability varies throughout the G-7. Although we find little predictability in U.S. revisions, the data revisions for several foreign countries are highly predictable. We also perform a simple real-time forecasting exercise showing that for several countries, the predictability of data revision could be used to generate improved preliminary data.

JEL codes: C22, C53, C82

Keywords: vintage data, preliminary data, final data, revision, GDP.

MACROECONOMIC DATA ARE OFTEN subject to large revisions after initial release and for many data series, the data revision process continues essentially indefinitely. The inaccuracy of initial data obviously complicates decision making by policymakers and other agents whose optimal choices depend on the state of the economy. Several authors have recently emphasized that the revision process also complicates the *ex post* analysis of macro data. Macroeconometric

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JON FAUST is affiliated with the International Finance Division, Federal Reserve Board (E-mail: faustj@frb.gov, <http://patriot.net/~faustj>). JOHN H. ROGERS is affiliated with the International Finance Division, Federal Reserve Board (E-mail: john.h.rogers@frb.gov). JONATHAN H. WRIGHT is affiliated with the International Finance Division, Federal Reserve Board (E-mail: jonathan.h.wright@frb.gov).

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work is generally based on the most fully revised available data (ignoring any earlier data vintages). The conclusions of such work rest on the implicit assumption that at each point in time, agents perfectly predict future data revisions. This assumption is particularly curious given the fact that revisions should be completely unpredictable if the statistical agency is efficiently processing available information. Studies of the revision process in U.S. GDP data show that revisions are unpredictable or only weakly predictable (e.g., Mankiw and Shapiro 1986).

Macroeconometric work based on real-time data often yields substantively different conclusions from work ignoring revisions (see, for example, Croushore and Stark, 2001, Evans, 1998, Orphanides, 2000, 2001, Christofferson, Ghysels, and Swanson, 2001). The predictive power of forecasting models has been found to be quite sensitive to whether the forecasts are based on real-time or revised data (see, for example, Diebold and Rudebusch, 1991, Faust, Rogers, and Wright, 2002, Robertson and Tallman, 1998).

In this paper, we study the forecastability of revisions to GDP announcements for the G-7 countries. We extend earlier work both by including a broader range of economies and by taking a longer sample of data—at longest 1965–97. We consider both “short-term” revisions, those that occur in the 24 months after the preliminary announcement, and “long-term” revisions, those that have taken place by the end of our sample.

Our results generally support earlier work finding little predictability of revisions in the U.S., but for several other countries we find a great deal of predictability. For Italy, Japan, and the UK, about half the variance of long-term revisions can be accounted for by information available at the time of the initial announcement. For Canada, Germany, Italy, Japan, and the UK, at least a quarter of the variance of short-term revisions can be accounted for by information known at the time of the preliminary announcement. Preliminary announcements in some countries are significantly biased. The variable with the most power to predict future revisions to GDP is the preliminary GDP announcement itself—extreme preliminary announcements tend to be revised toward the mean. We also consider six other predictors that are publicly known at the time of the preliminary announcement. Although statistically significant in some cases, these generally do not add much to the predictive power.

In many contexts, the in-sample fit of forecasting models proves to be a poor indicator of out-of-sample performance. In the current context, it is not clear that the predictability detected in forecast efficiency regressions could be exploited in real time to improve on preliminary estimates of GDP. As a crude check of the out-of-sample usefulness of our results, we perform a *real-time* forecasting exercise assessing. We find that for Canada, Germany, Italy, Japan, and the UK, using our models to adjust the preliminary data (using only coefficient estimates and data available at the time of the preliminary announcement) would lead to a statistically significant reduction of about one-quarter in the mean square error (MSE) of short-term revisions.

The plan of the remainder of this paper is as follows. Section 1 reviews standard models of the revision process, the role of efficient forecasts, and forecast tests. Section 2 describes our dataset. Sections 3 and 4 present baseline results and extensions, and the final section provides some interpretation of the results.

1. REVISIONS, NEWS, NOISE, AND EFFICIENT PREDICTIONS

The revision process is characterized by two polar cases labelled *news* and *noise* by Mankiw, Runkle, and Shapiro (1984) and Mankiw and Shapiro (1986). Under the *noise* view, the preliminary GDP data are polluted with measurement error—noise, that is uncorrelated with the true values. The preliminary GDP number will not be an optimal estimate of GDP in this case, and agents face a filtering problem in forming their optimal estimate. Various approaches to this filtering problem have been proposed (Howrey, 1978, 1984, De Jong, 1987, and Mariano and Tanizaki, 1995).

Under the *news* characterization, the statistical agency optimally uses all available information in forming the preliminary number, and revisions must reflect news that arrives after the announcement. The *news* view will be appropriate so long as the statistical bureau is choosing the preliminary number to minimize any one of a number of standard loss functions that are symmetric and increasing in the size of revisions. Some national statistical offices explicitly discuss minimizing revisions as a goal of their processes, and most presumably include this among the desiderata in data construction.¹ When revisions are minimized, the preliminary number will be what is known as a rational, efficient, or optimal forecast of the subsequently revised data. The revision will be orthogonal to information available when it is produced. More formally, under both the *news* and *noise* views, we can characterize the preliminary data as equal to the final plus an error term:

$$X_t^p = X_t^f + \varepsilon_t. \quad (1)$$

Under the noise view, ε_t is orthogonal to X_t^f , while in the news view, ε_t is orthogonal to X_t^p . Obviously, there are intermediate cases in which ε_t is correlated with both preliminary and final data.

Predictability of revisions can arise in many ways. Suppose output data are available before income and expenditure data so that the preliminary announcement gives GDP measured on an output basis only. The final data involve some reconciliation of output-based, expenditure-based, and income-based methods.² Even if each of the three sources of information gives an unbiased estimate of the final and each is similarly noisy, mechanically using the output-based number (or any of

1. For the U.S., see the editor's note in Mankiw and Shapiro (1986); for the UK, Barklem (2000); and for Japan, Economic Research Institute (2000).

2. The UK process is roughly like this hypothetical example. These three bases for GDP measurement and the reconciliation are explained in Reed (2000).

the three) as preliminary data without appropriate filtering will give rise to predictable revisions. If one only has the output data, an optimal linear estimate of GDP would require scaling the output-based number by a factor reflecting the signal-to-noise ratio in those data.

We follow others in attempting to distinguish the news and noise views using standard forecast efficiency tests. Under the news view, revisions must be mean zero; under *noise*, they need not be. Thus, we initially test the hypothesis that the revisions are unbiased. If ε_t in Equation (1) is correlated with X_t^p as in the noise case, X_t^p will predict the subsequent revision, $R(t) \equiv X_t^f - X_t^p$. To test this, we use the classic forecast efficiency test, which involves running the regression,

$$R_t = \alpha + \beta X_t^p + u_t. \quad (2)$$

We test the forecast efficiency implication that $\alpha = \beta = 0$. This is sometimes known as the Mincer–Zarnowitz test (Mincer and Zarnowitz 1969). We will use this as our baseline test for forecast rationality. One can also augment the set of explanatory variables in this regression with any variable known at time t . Forecast rationality implies that all the coefficients should be zero in any such regression.

Several earlier papers have applied these tests using revisions to money stock and/or output data, typically for the U.S.³ Mankiw and Shapiro (1986) find little evidence against the null hypothesis of forecast rationality using a short sample of U.S. GNP data and so characterize the data revision process as incorporating news. Using U.S. consumption data, Croushore and Stark (2001) find that revisions up to 1 year after the initial data release are uncorrelated with the preliminary data, but that subsequent revisions are weakly predictable using preliminary data. Barklem (2000) finds evidence of bias in revisions to various UK preliminary releases, including GDP.

Especially from the standpoint of assessing efficiency of the data construction agency, it seems clear that short-term revisions are of more interest than long-term revisions. Long-term revisions incorporate changes in definitions and the effects of changes in the base year. It would be reasonable for the statistical agency to strive to make the preliminary data the best possible estimate of GDP *under the current measurement system*. Furthermore, the conceptual and definitional changes in benchmark revisions are arguably one-time changes so that past predictability may be of little relevance to what we should expect going forward.

Setting aside the narrow question of the efficiency of the statistical agency, it is clear that long-term revisions may be of interest. If agents are interested in some *true* state of the economy, and if all revisions including definitional changes move the data toward a better measure of the true state, then the predictability of long-term revisions may be of most interest. In any case, we present both sets of results.

3. Mankiw, Runkle, and Shapiro (1984), using U.S. money stock data, reject the null hypothesis of forecast rationality. Other authors, including Kavajecz and Collins (1995), also using U.S. money stock data, reject forecast rationality when using seasonally adjusted data, but not when using the unadjusted data. Similar results are obtained for Canadian money stock data by Milbourne and Smith (1989).

All of this work, including ours, requires one caveat: because data construction methods are constantly being revised, it is unclear whether past predictability is evidence of future predictability. We partially address this problem by considering both a full sample and a more recent 10-year sample and by using out-of-sample methods. It remains true that methods in most countries have changed considerably, even in the past decade. Considering even shorter samples is problematic because statistical power falls as the sample size shrinks and because very recent data has not had much time to be revised.

2. THE DATA

Our data comprise preliminary estimates of real quarterly GDP growth rates and their subsequent revisions for the G-7 countries.⁴ The data come from the OECD's *Main Economic Indicators* (MEI). To obtain our *preliminary* estimate of GDP growth for a given quarter, we find the first monthly issue of the (hardcopy) MEI in which GDP is reported for the relevant quarter and we calculate the implied GDP growth rate.⁵ Throughout, the growth rate is defined as the quarter-over-quarter percent change (not annualized) of real seasonally adjusted GDP (GNP was in fact reported in early data vintages). We compute a short-term and a long-term revision of the preliminary growth rate. The long-term revision is the revision between the preliminary and the data as they stood in the most recent CD of MEI.⁶ We call these latter data our *final* data, in that they are the latest vintage that we consider. Of course, GDP data continue to be revised indefinitely, for example, for base year and definitional changes. To ensure that our *final* numbers have had at least some revisions, we end the sample for all countries in 1997Q4. The short-term revision is between the preliminary growth rate and the growth rate as it stood in the MEI 24 months after the preliminary number appeared.

For the United States, United Kingdom, and Canada, preliminary data are available beginning in 1965Q1. For Japan, the starting date is 1970Q1; for Italy and Germany, 1979Q4; and for France, 1987Q4. The German data refer to West Germany until 1994Q4 and to all of Germany in all subsequent quarters.

3. PREDICTABILITY OF GDP GROWTH REVISIONS

3.1 *Bias in Revisions and Other Summary Statistics*

Table 1 shows the mean revisions, the mean absolute revisions, and the root mean square revisions, for both short-term and long-term revisions. The table gives results

4. The data used in this paper are available from the authors.

5. Although MEI is published monthly, we are not guaranteed that the first number published in MEI is the first number ever released. Data for a given quarter is usually reported in MEI soon after the end of that quarter (usually within 2 or 3 months). For the United States, our preliminary data for GDP growth are nearly identical to that in the Croushore–Stark dataset (Croushore and Stark 1999). The Croushore–Stark data are obtained from the May, August, November, and February issues of the Survey of Current Business for quarters 1, 2, 3, and 4, respectively.

6. This is the April 2003 CD, except that this CD does not include data for the former West Germany, nor does the data on this CD cover the whole sample period for Canada or Japan. We, therefore, instead use the October 2000 CD of MEI for Canada, West Germany, and Japan only.

TABLE 1
SUMMARY OF REVISIONS AND ANNOUNCEMENTS FOR TWO SAMPLES

	Canada	France	Germany	Italy	Japan	UK	U.S.
<i>Short-Term Revisions, Sample Start–1997Q4</i>							
Mean	0.09	0.02	0.03	0.09	0.01	0.18	0.06
<i>t</i> -stat	1.69	0.34	0.26	1.85	0.27	2.76	1.98
Root mean square	0.66	0.32	1.45	0.62	0.62	0.97	0.35
Mean absolute	0.42	0.25	1.01	0.44	0.46	0.68	0.27
Prelim mean growth	0.69	0.48	0.37	0.37	0.98	0.27	0.65
<i>Short-Term Revisions, Sample 1988Q1–1997Q4</i>							
Mean	–0.01	0.00	0.08	0.03	–0.00	0.15	–0.01
<i>t</i> -stat	–0.20	0.08	0.71	0.56	–0.04	1.70	–0.13
Root mean square	0.26	0.31	1.14	0.43	0.60	0.64	0.30
Mean absolute	0.21	0.24	0.67	0.33	0.49	0.31	0.23
Prelim mean growth	0.52	0.49	0.37	0.42	0.61	0.30	0.60
<i>Long-Term Revisions, Sample Start–1997Q4</i>							
Mean	0.18	0.00	0.12	0.13	–0.07	0.31	0.15
<i>t</i> -stat	3.01	0.05	1.51	1.73	–0.99	4.65	3.89
Root mean square	0.81	0.35	0.94	0.86	0.84	1.15	0.55
Mean absolute	0.58	0.28	0.68	0.63	0.66	0.86	0.43
<i>Long-Term Revisions, Sample 1988Q1–1997Q4</i>							
Mean	0.02	–0.02	0.22	0.03	0.05	0.25	0.11
<i>t</i> -stat	0.38	–0.34	1.81	0.66	0.68	2.35	2.58
Root mean square	0.34	0.32	1.05	0.57	0.84	0.75	0.37
Mean absolute	0.27	0.26	0.72	0.42	0.67	0.44	0.31

NOTES: The growth rate numbers are quarter-over-quarter growth rates in percent. The long-term revision is the final value minus the preliminary; the short-term revision is the value from 24 months after the preliminary minus the preliminary. The *t*-statistics are based on autocorrelation and heteroskedasticity consistent standard errors and are for the hypothesis that the mean is zero. The sample ends in 1997Q4 for all countries. The full sample begins in 65Q1, 87Q4, 79Q4, 79Q4, 70Q1, 65Q1, and 65Q1, for the seven countries, respectively. The results for Germany should be viewed with caution because different databases were used to calculate preliminary and final growth rates.

over the longest span of data available for each country and for the more recent 10 years, 1988Q1–1997Q4, for all countries.

Revisions are quite large for all countries. The root mean squared error (RMSE) of short-term revisions in the full sample ranges from 0.32 percentage points (unannualized) for France to 1.45 percentage points for Germany. Long-term revisions are larger than short-term revisions for all countries except Germany. Revisions are generally smaller for the shorter, recent sample. In comparing the magnitudes of data revisions across countries, one should bear in mind that some countries, including the United Kingdom, issue their preliminary data much more quickly than others.⁷ For the full sample, mean short-term and long-term revisions are positive for all countries, except for Japan's long-term revision, indicating a general bias toward pessimism in the initial announcements. The bias is quite large in the UK at 0.31 percentage points (more than a full percentage point at an annualized rate) for long-term revisions and slightly more than half that for short-term revisions. We also report a *t*-statistic for testing the forecast efficiency hypothesis that the mean revision is

7. The Federal Reserve Board has recorded the dates of first announcements on GDP growth over the last 5 years, for the G-7 countries. The average time between the end of the quarter and the first announcement is 56 days across the G-7 as a whole, but is only 26 days for the United Kingdom and 30 days for the United States.

zero (standard errors here and throughout are heteroskedasticity and autocorrelation robust, using Newey–West standard errors with a lag truncation parameter of four (Newey and West 1987). The results are not very sensitive to the choice of this lag parameter). The hypothesis that the mean short-term revision is zero is rejected for the U.S. and the UK (unless otherwise stated, we use the 5% level to judge significance and test rejections). The mean bias in the long-term revisions is also significantly positive for Canada, the U.S., and the UK. Statistically significant bias for the U.S. can also be found in the Croushore–Stark data.⁸ Bias is, of course, the simplest form of predictability in revision; we now examine the predictability of the data revision process more comprehensively.

3.2 Forecastable Revisions: An Informal Look

If the data revision process only incorporates news, then there should be no systematic relation between the preliminary announcement and the subsequent revision. In fact, one can see a clear negative relationship in the scatter plots of short-term revisions against the preliminary growth rate for several countries, notably Germany, Japan, Italy, and the UK (Figure 1). For these countries, high preliminary numbers are systematically revised downwards and low preliminary numbers are systematically revised upwards. The same relationship is apparent in the scatter plot of the long-term revisions against the preliminary data (Figure 2).

This informal evidence against forecast rationality is exactly as one would expect under the *noise* interpretation of the data revision process. The noise contributes to excessive variance of the preliminary data. Further, unusual observations in the preliminary data tend to be revised toward more normal values as the noise is removed. These informal results are substantiated in the following sections.

3.3 The Baseline Forecast Efficiency Regression

First, we report evidence from our baseline tests of forecast rationality—tests of the hypothesis that $\alpha = \beta = 0$ in the regression (2). For short-term revisions, the *F*-statistic for testing this hypothesis rejects forecast rationality over the full sample for every country except France and the U.S. (Table 2, first panel). In the recent sample, rationality is rejected only for Japan (second panel). For long-term revisions in the full sample, we find rejections of rationality for all countries but France, which has the shortest sample (third panel). Overwhelming rejections are obtained for Japan, Italy, and the United Kingdom. The degree of predictability in data revisions varies substantially across countries. The adjusted R^2 ranges from 1% for France to 63% for Italy. In the recent 40 quarter sample (Table 2, fourth panel), we reject forecast rationality for long-term revisions for all countries except France and Canada. The explanatory power of the regression remains substantial for Italy,

8. In calculations not shown in this paper, we find that in the Croushore–Stark data, available on the website of the Philadelphia Federal Reserve Bank, the revision of output growth going from the first release to the most recently available data is significantly positive, as is the revision going from the first release to the second release (one-quarter later).

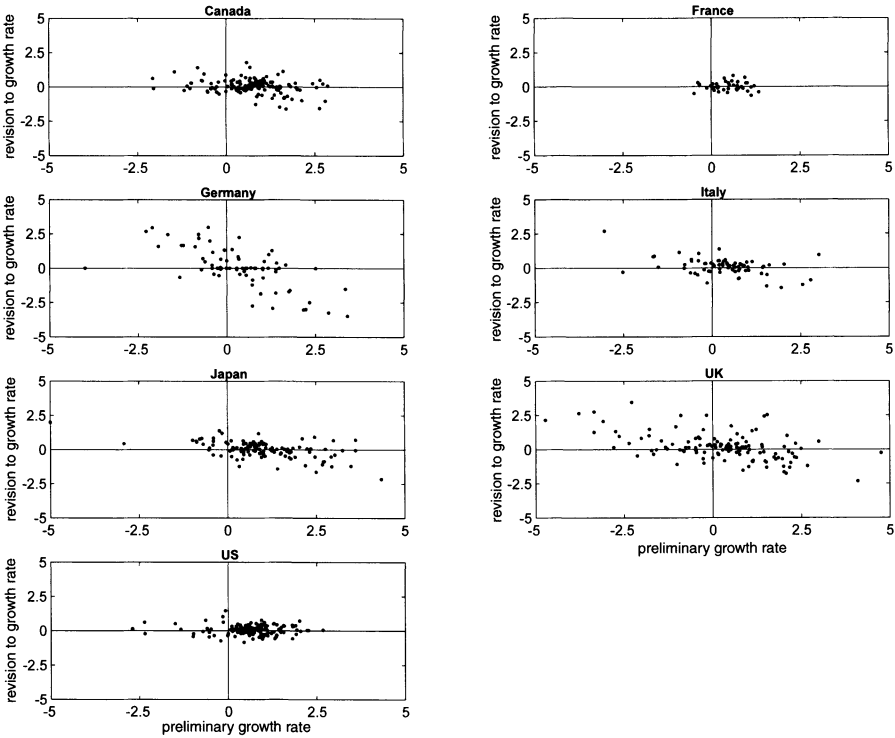


FIG. 1. Preliminary GDP Growth Rates and Short-Term Revisions

Germany, Japan, and the UK. The finding that short-term revisions are not significantly predictable in the U.S. is consistent with existing literature (e.g., Mankiw and Shapiro 1986).

Overall, we take these results as evidence of considerable predictability of GDP revisions in some countries in our sample. Predictability exists both in the full and recent samples. Even short-term revisions are predictable, though on the whole less predictable than long-term ones. We now turn to some robustness checks of and extensions to these basic results. To preview, neither the robustness checks nor the extensions alter much the basic conclusions just illustrated.

3.4 Robustness Checks

Especially in the early years of the sample, many of the preliminary GDP growth rates are quite extreme. This is particularly true for the United Kingdom in the 1970s. For example, the preliminary growth rate for the United Kingdom in 1971Q1 was -4.7% (not annualized): this was subsequently revised to -1.2% . To see if our results are driven by such outliers, we re-ran the regressions in Table 2, deleting

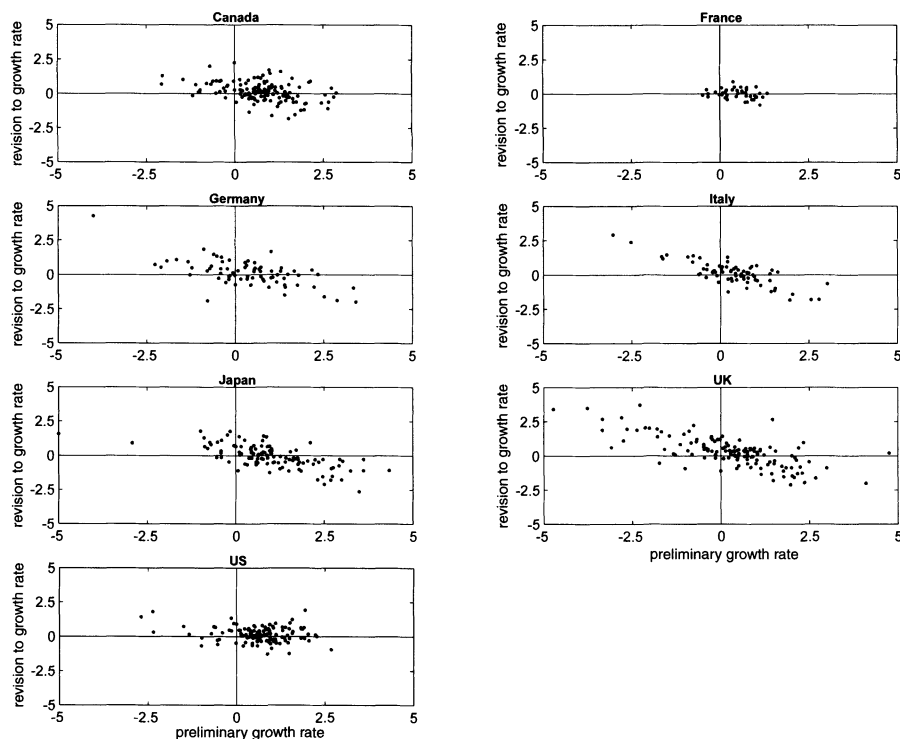


FIG. 2. Preliminary GDP Growth Rates and Long-Term Revisions

all observations for which the preliminary growth rate deviated from the country-mean by more than 3 percentage points.⁹ The results (Table 3) are entirely consistent with the earlier results (Table 2): these outliers are not driving the results.

As another robustness check, we reran the forecast rationality regressions using three exact nonparametric tests, as proposed by Campbell and Dufour (1995). The results are quite similar to the results in Table 2, but are not shown to conserve space; they are available from the authors.

3.5 A Real-Time Forecasting Exercise

Our estimates above are based on the entire sample period, and could not have been used during the sample period by statistical agencies to generate better preliminary announcements. As a simple check of whether statistical agencies could have done better with data available at the time of initial release, we undertake a real-time forecasting exercise in which we construct a recursively-adjusted preliminary growth rate series.

9. There were nine such outliers in the United Kingdom, three in Japan, two each in Germany, Italy, and the United States, one in Canada, and none in France. All but two of these outliers occurred prior to 1987.

TABLE 2
BASELINE FORECAST EFFICIENCY REGRESSION

	Canada	France	Germany	Italy	Japan	UK	U.S.
<i>Short-Term Revisions, Full Sample</i>							
Constant	0.29	0.02	0.31	0.19	0.26	0.27	0.07
	3.00	0.26	2.15	2.72	4.46	4.26	1.68
Prelim	-0.30	0.01	-0.76	-0.27	-0.25	-0.32	-0.01
	-2.97	0.07	-4.15	-2.49	-4.94	-6.15	-0.37
F	9.50	0.10	18.50	7.90	25.80	43.00	3.90
p-val	(0.01)	(0.94)	(0.00)	(0.02)	(0.00)	(0.00)	(0.14)
R^2_U	0.25	0.00	0.45	0.23	0.28	0.30	0.03
$\bar{R}^2$	0.23	-0.03	0.44	0.20	0.27	0.26	-0.01
<i>Short-Term Revisions, 1988Q1-1997Q4</i>							
Constant	-0.04	-0.00	0.26	0.14	0.17	0.31	0.02
	-0.97	-0.04	1.69	1.54	2.08	1.68	0.18
Prelim	0.05	0.01	-0.49	-0.28	-0.29	-0.54	-0.04
	0.84	0.15	-2.37	-1.89	-4.02	-1.73	-0.35
F	1.30	0.00	5.80	3.60	16.70	3.00	0.20
p-val	(0.52)	(0.99)	(0.06)	(0.16)	(0.00)	(0.22)	(0.92)
R^2_U	0.01	0.00	0.33	0.18	0.33	0.44	0.00
$\bar{R}^2$	-0.01	-0.03	0.31	0.16	0.32	0.40	0.02
<i>Long-Term Revisions, Full Sample</i>							
Constant	0.44	0.07	0.30	0.37	0.35	0.45	0.23
	4.94	0.93	3.05	5.99	4.70	6.28	3.19
Prelim	-0.38	-0.14	-0.47	-0.65	-0.43	-0.53	-0.12
	-4.77	-1.22	-4.62	-10.52	-7.90	-8.66	-1.36
F	26.5	1.6	21.4	119.1	67.7	86.1	17.0
p-val	(0.00)	(0.44)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
R^2_U	0.31	0.03	0.42	0.64	0.45	0.56	0.11
$\bar{R}^2$	0.27	0.01	0.41	0.63	0.44	0.52	0.03
<i>Long-Term Revisions, 1988Q1-1997Q4</i>							
Constant	0.06	0.04	0.44	0.21	0.32	0.44	0.17
	1.04	0.68	3.06	2.17	2.94	2.38	1.82
Prelim	-0.07	-0.13	-0.59	-0.42	-0.44	-0.66	-0.09
	-0.94	-1.14	-4.60	-2.83	-5.33	-2.33	-0.94
F	1.4	1.3	22.1	8.0	28.4	6.0	6.3
p-val	(0.49)	(0.52)	(0.00)	(0.02)	(0.00)	(0.05)	(0.04)
R^2_U	0.02	0.04	0.61	0.24	0.41	0.53	0.10
$\bar{R}^2$	-0.01	0.01	0.58	0.21	0.39	0.45	-0.01

NOTES: See the notes to Table 1. In each specification, the dependent variable is the revision in percent; independent variables are listed in column 1. The number in italics underneath each coefficient estimate is a *t*-statistic for the hypothesis that the coefficient is zero. The row labelled *F* is an *F*-test of the hypothesis that all coefficients are zero; the *p*-value for this test is given in the next row. The rows labelled R^2_U and $\bar{R}^2$ are the uncentered and adjusted centered *R*-squared, respectively.

For each quarter, starting in the 20th quarter of the full sample for each country, we run a regression of the short-term revision on a constant and the preliminary growth rate, using only data that was available in real time. Next, we adjust the preliminary data for that quarter by adding to it the fitted revision for that quarter implied by the regression. France is excluded from this exercise due to the short sample.¹⁰ We report the MSE of the unadjusted preliminary data and the recursively adjusted preliminary

10. It is important to emphasize that we consider only short-term revisions, not long-term revisions. We could not do long-term revisions, since the final growth rate is not known until the end of the sample, and so could not be used in a real-time experiment of this sort.

TABLE 3

BASELINE FORECAST EFFICIENCY REGRESSION, FULL SAMPLE, OBSERVATIONS WITH EXTREME PRELIMINARY GROWTH RATES DELETED

	Canada	France	Germany	Italy	Japan	UK	U.S.
<i>Short-Term Revisions</i>							
Constant	0.17	0.02	0.42	0.13	0.23	0.22	0.05
	3.13	0.26	3.20	1.88	3.28	2.97	1.22
Prelim	-0.16	0.01	-0.88	-0.25	-0.21	-0.26	0.01
	-2.76	0.07	-7.55	-3.19	-3.21	-3.23	0.26
<i>F</i>	10.3	0.10	57.20	10.30	11.30	12.00	3.70
<i>p</i> -val	(0.01)	(0.94)	(0.00)	(0.01)	(0.00)	(0.00)	(0.16)
R^2_{adj}	0.09	0.00	0.50	0.19	0.17	0.14	0.02
$\bar{R}^2$	0.07	(0.03)	0.49	0.18	0.16	0.12	-0.01
<i>Long-Term Revisions</i>							
Constant	0.35	0.07	0.23	0.35	0.45	0.44	0.15
	4.48	0.93	2.93	5.42	5.48	5.65	3.91
Prelim	-0.28	-0.14	-0.35	-0.66	-0.51	-0.56	-0.03
	-4.49	-1.22	-5.29	-10.02	-8.19	-7.17	-0.56
<i>F</i>	22.7	1.6	28.6	104.9	69.4	54.8	20.2
<i>p</i> -val	(0.00)	(0.44)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
R^2_{adj}	0.18	0.03	0.27	0.60	0.45	0.48	0.07
$\bar{R}^2$	0.14	0.01	0.25	0.59	0.44	0.45	-0.01

NOTES: See the notes to Table 2. The table reports the same results as Table 2, except that observations with outlier preliminary data (as defined in the text) are deleted.

data as forecasts of the after-24-months data (Table 4). If the revisions are not forecastable, we would expect the MSE of the adjusted preliminary data to be higher—fitting irrelevant parameters *raises* out-of-sample mean square prediction error. If the model has substantial predictive power, we would obviously expect the MSE of the adjusted data to be lower. We report a Diebold–Mariano (Diebold and Mariano 1995) test of the hypothesis that these MSE are equal versus the alternative that the adjusted data have smaller MSE along with the associated *p*-values constructed following Clark and McCracken (2002). The construction of these *p*-values is complicated because the models are nested and so the difference in MSE under the null comes exclusively from parameter estimation error, but

TABLE 4

REAL-TIME FORECASTS: MSE OF PRELIMINARY DATA AND REAL-TIME ADJUSTED PRELIMINARY DATA AS FORECASTS OF DATA AFTER 24 MONTHS

	Canada	Germany	Italy	Japan	UK	U.S.
MSE (Raw)	0.45	2.05	0.24	0.23	0.99	0.14
MSE (Adjusted)	0.37	1.43	0.18	0.17	0.72	0.15
Diebold–Mariano	0.74	0.87	1.37	1.79	2.32	-1.08
<i>p</i> -val	(0.02)	(0.02)	(0.00)	(0.00)	(0.00)	(0.49)

NOTES: This table shows the MSE of the raw preliminary data and of the recursively adjusted preliminary data (adjusted using Equation (2), as discussed in the text), as forecasts of the data as observed after 24 months. The row labelled Diebold–Mariano is the Diebold–Mariano statistic (MSE-*t* in the notation of Clark and McCracken 2002) measuring the difference between MSE (Raw) and MSE (Adjusted); the *p*-values for this statistic, constructed following Clark and McCracken as described in the appendix, are contained in the last row. The test is a one-sided test, rejecting if the observed statistic is greater than the critical value.

correct p -values can be obtained from the work of Clark and McCracken (2002), which extends earlier work of McCracken (1999). The test and the construction of p -values are described briefly in the appendix.

For all of the countries except the United States, the MSE of the real-time recursively-adjusted preliminary data is smaller than that of the original preliminary data, and this reduction in MSE is statistically significant at the 5% level, using the Diebold–Mariano test. Leaving aside the U.S., the procedure reduces the short-term revision MSE by about one-quarter.

4. AUGMENTED FORECAST EFFICIENCY REGRESSIONS

Forecast rationality requires that the revisions be unforecastable using any data that was known at the time the preliminary data were released. To see if added explanatory variables strengthen the evidence against forecast rationality, we augment the basic regression with seasonal dummies and five variables known at the time that the preliminary data were released: lagged preliminary data, the growth rate of equity prices, a 3-month interest rate, oil price inflation, and a dummy variable for national elections (Canada, Japan, UK, and U.S. only).¹¹ We include the seasonal dummies and lagged preliminary data to check for additional simple forms of dynamics in the original specification. We follow Mankiw and Shapiro (1986) in using equity prices and short-term interest rates as business cycle indicators. We add oil prices for the same reason. The business cycle variables may have predictive power if the systematic inefficiencies in the data construction process are affected by the state of the business cycle.

We include the election dummies to capture a novel variant on the Nordhaus-style (1975) political business cycle. Nordhaus suggested that incumbent governments might attempt to boost the economy before elections to enhance election prospects. Incumbents might prefer to artificially boost the economic data, rather than actually stimulating the economy. Simply manipulating the data avoids some of the efficiency costs of deviating from (otherwise) optimal policy, and the evidence would disappear from the historical record as the data are revised. There have been accusations in the press of this sort of manipulation in Japan (New York Times 2000).

To investigate this possibility, we create an election dummy variable that is one if the preliminary number is the last one announced before a national election and zero otherwise. We include this dummy variable only for Canada, Japan, the United Kingdom, and the United States because for the other countries there were not enough elections within our sample period to obtain meaningful results.¹² For the

11. Equity prices are measured by the stock price indices for each country reported in MEI. The 3-month interest rate for each country is also taken from MEI. The oil price is the spot West Texas intermediate crude price.

12. Within our sample period, there were 8 elections in the United States, 9 in the United Kingdom, 10 in Canada, and 16 in Japan, but only 3 in France and 5 each in Germany and Italy. In the United States, the elections we refer to are presidential elections (midterm elections were disregarded). For Canada, Japan, and the United Kingdom, they are all national parliamentary elections (in either house, in the case of Japan). We are grateful to Deepak Mishra for providing us with some election data.

four relevant countries, we construct the dummy variable by obtaining the exact dates of elections and of the preceding initial GDP announcements from various national newspapers. If the data for the election quarters are optimistic, then we expect revisions for these quarters to be systematically more negative than at other times.

4.1 Results for the Augmented Regressions

When we include all the additional regressors (Table 5), forecast rationality is rejected for all countries except the U.S. for short-term revisions, and is rejected for all seven countries for long-term revisions. This contrasts with five and six rejections for short-term and long-term revisions, respectively, reported in Table 2.

In the regressions with short-term revisions, the interest rate variable is statistically significant for France and Germany and the stock return is significant for Japan. The election variable is never significant. The additional regressors do not, however, substantially increase the adjusted R^2 s of the regressions.

In the regressions with long-term revisions, each regressor is statistically significant, at least at the 10% level, for at least one country. For the UK and France, oil price inflation is significant; for the UK and Japan, the interest rate is significant; for Japan, equity price growth is significant at the 10% level. The seasonal dummies are jointly significantly different from zero for the UK and France, but for these countries alone. Only in the case of France, however, is there an appreciable increase in explanatory power of the regression relative to the basic specification.

For Japan alone, the election dummy is a marginally significant (significant at the 10% level) predictor of the long-term revision. The estimated election effect is large, as the last GDP growth rate announced before an election is subsequently revised downwards by about 0.3 percentage points more than for other quarters (thus, the revision is over a full percentage point greater at an annualized rate).

We also ran the regressions in Table 5 using only the subsample of data from 1988Q1 on, but omitting the election dummy (because there are too few elections in this subsample). The results were similar to those in Table 5, and are not shown, so as to conserve space, but are available from the authors on request: the null of forecast efficiency was rejected for all countries for both short-term and long-term revisions.

5. CONCLUSIONS

Revisions to GDP announcements are quite large in all the G-7 countries. We find only very weak evidence that U.S. GDP revisions are predictable, but find substantially more evidence of predictability abroad. There is predictability in short-term and long-term revisions, though the evidence is stronger for the long-term revisions. We find that real-time recursive adjustment of preliminary data could have significantly reduced the MSE of revisions over the subsequent 24 months in

TABLE 5

FORECAST EFFICIENCY REGRESSION, SHORT- AND LONG-TERM REVISIONS, FULL SAMPLE

	Canada	France	Germany	Italy	Japan	UK	U.S.
<i>Short-Term Revisions</i>							
Constant	0.44	0.44	0.86	-0.53	0.60	0.13	0.05
	2.00	1.31	1.70	-0.82	2.90	0.63	0.38
Prelim	-0.36	0.03	-0.78	-0.35	-0.27	-0.31	-0.00
	-3.80	0.45	-4.04	-3.87	-5.63	-6.03	-0.02
Prelim(-1)	0.08	0.25	0.02	0.04	0.02	-0.02	-0.04
	1.59	1.70	0.16	0.68	0.40	-0.37	-0.70
Q1	-0.02	-0.02	0.10	-0.11	-0.11	0.49	0.07
	-0.12	-0.08	0.26	-0.70	-0.74	2.05	0.80
Q2	-0.04	-0.11	0.29	-0.29	0.16	0.28	0.09
	-0.40	-0.96	0.92	-1.99	1.09	1.51	1.16
Q3	0.03	-0.17	0.28	-0.11	-0.01	0.39	0.01
	0.14	-1.46	0.67	-0.48	-0.11	2.32	0.08
Oil	-0.73	-0.63	0.73	0.27	0.24	-0.00	-0.13
	-1.22	-1.21	0.33	0.29	0.31	-0.00	-0.18
Stock ret	0.02	-0.11	-0.13	0.55	-0.40	0.00	0.01
	1.55	-0.61	-0.47	1.58	-2.18	0.24	1.08
Interest	-0.02	-0.05	-0.09	0.03	-0.04	-0.02	-0.00
	-1.57	-3.31	-2.03	1.06	-1.40	-0.92	-0.20
Election	0.01	—	—	—	0.12	0.20	-0.04
	0.08	—	—	—	1.07	0.82	-0.42
F	30.50	63.00	83.10	50.30	88.20	102.80	14.50
p-val	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.15)
R^2_{Uj}	0.32	0.31	0.49	0.36	0.36	0.33	0.07
$\bar{R}^2$	0.26	0.14	0.43	0.27	0.30	0.25	-0.02
F* (seas)	0.40	13.16	0.84	5.88	3.77	7.11	2.28
p-val	(0.94)	(0.00)	(0.84)	(0.12)	(0.29)	(0.07)	(0.52)
<i>Long-Term Revisions</i>							
Constant	0.53	0.43	-0.03	0.77	0.90	0.58	0.23
	2.69	1.38	-0.08	1.47	3.65	2.86	1.18
Prelim	-0.46	-0.01	-0.44	-0.71	-0.46	-0.51	-0.10
	-6.32	-0.24	-3.87	-10.43	-8.68	-9.46	-1.18
Prelim(-1)	0.17	0.23	0.08	-0.01	0.02	0.03	-0.03
	2.31	3.73	1.42	-0.28	0.47	0.53	-0.59
Q1	-0.12	-0.10	-0.29	0.02	-0.14	0.57	0.16
	-0.73	-0.66	-0.94	0.12	-0.71	2.95	1.15
Q2	0.11	-0.13	-0.06	-0.04	0.13	0.48	0.11
	0.70	-1.39	-0.37	-0.21	0.82	2.56	0.96
Q3	-0.19	-0.32	-0.20	-0.09	0.03	0.40	-0.17
	-1.17	-3.12	-1.07	-0.60	0.20	2.55	-1.07
Oil	0.27	-1.87	-1.64	-0.86	-0.36	-2.80	0.12
	0.30	-2.84	-1.14	-0.61	-0.37	-3.01	0.24
Stock ret	0.01	-0.26	0.31	-0.08	-0.43	0.00	0.01
	0.84	-1.32	1.15	-0.31	-1.98	0.22	1.11
Interest	-0.01	-0.02	0.03	-0.02	-0.07	-0.05	-0.00
	-1.24	-1.21	0.71	-0.90	-2.16	-2.50	-0.21
Election	-0.05	—	—	—	-0.28	-0.04	-0.16
	-0.24	—	—	—	-1.67	-0.16	-1.03
F	76.9	91.6	43.7	168.5	130.8	576.5	38.9
p-val	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
R^2_{Uj}	0.40	0.34	0.47	0.67	0.53	0.65	0.20
$\bar{R}^2$	0.33	0.17	0.39	0.62	0.48	0.59	0.07
F* (seas)	5.80	14.5	1.54	0.53	1.69	10.93	7.09
p-val	(0.12)	(0.00)	(0.67)	(0.91)	(0.64)	(0.01)	(0.07)

NOTES: See the notes to Table 2. The row labelled F* (seas) is for a joint test of the joint hypothesis that the three seasonal dummy variables are zero in the regression.

all of the G-7 countries except the U.S. (where there is little predictability) and France (where the sample was too short to make an estimate).

When we find predictability of revisions, it is mostly due to the predictive power of the preliminary number: extreme values, large or small, in the preliminary growth rate tend to be revised toward the mean. This is exactly what one would expect under the *noise* view of revisions in which revisions remove measurement error from earlier announcements.

Forecastability of revisions is important for how we interpret preliminary data. It also has potentially important implications for macroeconometric work. Several recent papers have pointed out the problems with using fully revised data; such work rests on the assumption that agents perfectly anticipate revisions. A corrective that has been used is to re-do the work using preliminary data. Of course, this may rest on the assumption that revisions are completely unpredictable. This assumption is a reasonable approximation for the U.S., but for Italy, Japan, and the UK, revisions are quite predictable and neither extreme assumption is appropriate.

APPENDIX

In this appendix, we outline the procedure for testing the hypothesis that the two MSE in Table 4 are equal. If the preliminary data are used as a predictor of the final data, the prediction error is R_t . If the recursively adjusted data are used, the prediction error is $\hat{u}_t = R_t - \hat{\alpha}_t - \hat{\beta}_t' X_t^p$, where $\hat{\alpha}_t$ and $\hat{\beta}_t$ denote the recursive estimates of α and β , in the notation of Equation (2). Let p denote the number of observations on which the first out-of-sample forecast is based, P denote the number of observations used for out-of-sample prediction, and T denote the total sample size. The Diebold–Mariano statistic is $DM = P^{-1/2} \Sigma_{t=p}^T (R_t^2 - \hat{u}_t^2) / \hat{\sigma}$, where $\hat{\sigma}^2$ denotes the Newey–West estimator of 2π times the zero-frequency spectral density of $R_t^2 - \hat{u}_t^2$. The recursive out-of-sample forecast errors $\hat{u}_t$ are based on a model that has estimated parameters and that nests the model generating the forecast errors R_t . Under these conditions, the null asymptotic distribution of DM is not standard normal, but Clark and McCracken (2002) show that if $\frac{P}{p} \rightarrow \frac{1-\lambda}{\lambda}$, for fixed λ , the null limiting distribution of DM is $[\int_{\lambda}^1 s^{-1} W(s)' dW(s) - \frac{1}{2} \int_{\lambda}^1 s^{-2} W(s)' \Sigma W(s) ds] / [\int_{\lambda}^1 s^{-2} W(s)' W(s) ds]^{1/2}$, where $W(s)$ is a Brownian motion with covariance matrix $\Sigma = E(u_t^2)^{-1} E(\tilde{x}_t \tilde{x}_t')^{-1/2} \Gamma E(\tilde{x}_t \tilde{x}_t')^{-1/2}$, $\tilde{x}_t = (1, X_t^p)'$ is the vector of regressors in Equation (2) and Γ is 2π times the zero-frequency spectral density of $\tilde{x}_t u_t$. In general, this limiting distribution depends on the nuisance parameter Σ , but this can be consistently estimated in the usual way, using a Newey–West estimator of Γ . The asymptotic distribution can then be simulated to obtain p -values. Of course, each dataset will have a different estimate of Σ and so will require simulation of a different set of p -values. The test is a one-sided test, which rejects the null that the MSE are equal if the Diebold–Mariano statistic exceeds the critical value. In all the simulations used to construct p -values for Table 4, the mean of this null limiting

distribution was negative, which makes sense since overfitting raises out-of-sample mean square prediction errors.

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